On the Decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$: Quasimodels Meet the Patchwork Property

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Abstract

The description logics $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ extend $\mathcal{ALCI}$ with role boxes derived from the composition tables of the Region Connection Calculi RCC5 and RCC8. These logics were introduced by Wessel [10], who established PSPACE-hardness of $\mathcal{ALCI}_{RCC5}$ and EXPTIME-hardness of $\mathcal{ALCI}_{RCC8}$ but left decidability open. We settle both questions positively: concept satisfiability in $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ is decidable, with an EXPTIME upper bound. We give two decision procedures: a *type-elimination algorithm* and a *tableau calculus with blocking*. Both exploit the *patchwork property* of the RCC calculi—the fact that path-consistent atomic constraint networks are globally consistent [8]—to lift local (triple-wise) consistency to global model existence. For $\mathcal{ALCI}_{RCC8}$ this yields EXPTIME-completeness.

Keywords: description logic, spatial reasoning, Region Connection Calculus, decidability, tableau, quasimodel, patchwork property

1 Introduction

Description logics (DLs) are a family of knowledge representation formalisms that underpin ontology languages such as OWL [1]. The *standard* decision procedures for DLs exploit the *tree model property*: every satisfiable concept has a tree-shaped model, so one can search for a finite abstraction of such a tree (a "type elimination" or "tableau" argument).

The $\mathcal{ALCI}_{RCC}$ family, introduced by Wessel [10, 9], takes a different path. Here the roles are not abstract binary relations but the *base relations* of a qualitative spatial calculus—specifically, a Region Connection Calculus (RCC) [6]. Interpretations are constrained to be complete graphs in which every pair of domain elements is related by exactly one base relation, and the composition of any two base relations along a path is required to conform to the RCC composition table.

Formally, let $\mathcal{T}$ be one of RCC1, RCC2, RCC3, RCC5, or RCC8. The logic $\mathcal{ALCI}_{RCC}[\mathcal{T}]$—or simply $\mathcal{ALCI}_{RCCk}$ for the standard RCC-k variants—is the extension of $\mathcal{ALCI}$ (the basic DL $\mathcal{ALC}$ enriched with inverse roles) whose role names are the base relations of $\mathcal{T}$, and whose interpretations are subject to the *one-cluster, JEPD, composition table, and converse constraints* (Section 2).

Wessel [10] classified most of the family:

- $\mathcal{ALCI}_{RCC1}$ is NP-complete (equivalent to the modal logic S5).
- $\mathcal{ALCI}_{RCC2}$ is decidable (reduction to a two-variable fragment).
- $\mathcal{ALCI}_{RCC3}$ is decidable (reduction to the two-variable fragment of first-order logic with equality, $\mathrm{FO}^2[=]$).
- $\mathcal{ALCI}_{RCC5}$ is PSPACE-hard; decidability was left open.

*Claude is an AI assistant by Anthropic. This research was prompted by Michael Wessel (miacwess@gmail.com), who introduced the $\mathcal{ALCI}_{RCC}$ family in his doctoral work at the University of Hamburg and left the decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ as open problems [10, 9].

- $\mathcal{ALCI}_{RCC8}$ is ExpTime-hard; decidability was left open.

The difficulty with $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ stems from the combination of three properties: models are complete graphs (no tree model property), infinite models are required (no finite model property), and the composition tables have non-deterministic entries (the role boxes are not "admissible" in the sense of $\mathcal{ALCRA}_{SG}$ [9]).

Contribution. We prove that concept satisfiability in $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ is decidable, with an ExpTime upper bound. Combined with the known lower bounds, this yields ExpTime-completeness for $\mathcal{ALCI}_{RCC8}$. For $\mathcal{ALCI}_{RCC5}$, the exact complexity remains between PSpace and ExpTime.

Proof techniques. We develop two complementary decision procedures. First, a *quasi-model method* (Sections 4–6) that abstracts a (possibly infinite) complete-graph model into a finite structure consisting of *types*, *pair-types*, and *triple-types*, and a type-elimination algorithm deciding quasimodel existence. Second, a *tableau calculus* (Section 7) that constructs a finite completion graph by expanding individual nodes—analogous to standard DL tableaux, but adapted to complete-graph semantics. In the tableau, each new node is assigned a role to *every* existing node, and a blocking condition based on type equality ensures termination. The key technical ingredient in both procedures is the *patchwork property* of RCC5/RCC8 [8, 7]: path-consistent atomic constraint networks are globally consistent.

Related work. Lutz and Wolter [5] studied modal logics with RCC8 topological modalities interpreted over actual topological spaces; in that setting, reasoning is generally undecidable. Lutz and Miličić [4] showed that $\mathcal{ALC}$ extended with ω-admissible concrete domains (which include RCC5 and RCC8) is decidable, but the concrete-domain approach does not allow quantification *over* spatial roles (the defining feature of $\mathcal{ALCI}_{RCC}$). Borgwardt et al. [2] further refined the complexity of concrete-domain reasoning. The $\mathcal{ALCRA}_{SG}$ fragment [9] handles *admissible* role boxes (deterministic, functional, complete, associative) via reduction to $\mathcal{ALC}$ with general TBoxes, but the RCC5/RCC8 role boxes are non-deterministic and hence fall outside $\mathcal{ALCRA}_{SG}$.

2 Preliminaries

2.1 The Description Logic $\mathcal{ALCI}_{RCC5}$

Definition 2.1 (Syntax)**.** Let N_C be a set of *concept names* and let $N_R = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ be the set of *RCC5 base relations*, serving as role names. *Concepts* are built by the grammar:

$$C, D \ ::= \ A \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C \mid \exists R^{-}.C \mid \forall R^{-}.C$$

where $A \in N_C$ and $R \in N_R$. The *inverse* mapping on roles is $\mathsf{inv}(\mathsf{PP}) = \mathsf{PPI}$, $\mathsf{inv}(\mathsf{PPI}) = \mathsf{PP}$, $\mathsf{inv}(\mathsf{DR}) = \mathsf{DR}$, $\mathsf{inv}(\mathsf{PO}) = \mathsf{PO}$, $\mathsf{inv}(\mathsf{EQ}) = \mathsf{EQ}$.

We assume concepts are in *negation normal form* (NNF): negation is pushed to concept names. Every $\mathcal{ALCI}_{RCC5}$ concept can be converted to NNF in linear time. In NNF, inverse roles R^{-} can be absorbed: $\exists \mathsf{PP}^{-}.C = \exists \mathsf{PPI}.C$, etc. We therefore write role terms simply as elements of N_R.

Definition 2.2 (Semantics)**.** An $\mathcal{ALCI}_{RCC5}$ *interpretation* is a structure $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ where $\Delta^{\mathcal{I}} \neq \emptyset$ and $\cdot^{\mathcal{I}}$ assigns to each concept name A a set $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$ and to each base relation R a binary relation $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$, subject to:

(S1) **One cluster.** For all $x, y \in \Delta^{\mathcal{I}}$, there exists $R \in N_R$ with $(x, y) \in R^{\mathcal{I}}$.

(S2) **JEPD.** For distinct $R, S \in N_R$, $R^{\mathcal{I}} \cap S^{\mathcal{I}} = \emptyset$.

(S3) **Composition.** For all $R, S \in N_R$: $R^{\mathcal{I}} \circ S^{\mathcal{I}} \subseteq \bigcup_{T \in \mathsf{comp}(R,S)} T^{\mathcal{I}}$, where comp is the RCC5 composition table (Table 1).

(S4) **Converse.** $\mathsf{PPI}^{\mathcal{I}} = (\mathsf{PP}^{\mathcal{I}})^{-1}$, $\mathsf{DR}^{\mathcal{I}} = (\mathsf{DR}^{\mathcal{I}})^{-1}$, $\mathsf{PO}^{\mathcal{I}} = (\mathsf{PO}^{\mathcal{I}})^{-1}$.

Table 1: The RCC5 composition table. Entry for row S, column R gives $\mathsf{comp}(R, S)$, i.e., the possible relations for (a, c) when $R(a, b)$ and $S(b, c)$. The symbol $*$ denotes N_R, i.e., all five relations are possible.

comp	$\mathsf{DR}(a,b)$	$\mathsf{PO}(a,b)$	$\mathsf{EQ}(a,b)$	$\mathsf{PPI}(a,b)$	$\mathsf{PP}(a,b)$
$\mathsf{DR}(b,c)$	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
$\mathsf{PO}(b,c)$	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
$\mathsf{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathsf{PP}(b,c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$\mathsf{PPI}(b,c)$	DR	DR, PO, PPI	PPI	PPI	$*$

(S5) **Strong EQ.** $\mathsf{EQ}^{\mathcal{I}} = \{(x, x) \mid x \in \Delta^{\mathcal{I}}\}$.

Concept constructors are interpreted standardly: $(\exists R.C)^{\mathcal{I}} = \{x \mid \exists y.\, (x, y) \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}$, etc.

A concept C is *satisfiable* if there exists an interpretation $\mathcal{I}$ and $x \in \Delta^{\mathcal{I}}$ with $x \in C^{\mathcal{I}}$.

Remark 2.3 (Weak EQ semantics)**.** The variant with *weak* EQ (only $Id_{\Delta^{\mathcal{I}}} \subseteq \mathsf{EQ}^{\mathcal{I}}$, so EQ may also hold between distinct elements) reduces to the strong variant by adding the axiom $\top \sqsubseteq \forall \mathsf{EQ}.C \sqcap \forall \mathsf{EQ}^{-}.C$ for every relevant concept C and internalizing via the universal role $R_* = \mathsf{DR} \sqcup \mathsf{PO} \sqcup \mathsf{EQ} \sqcup \mathsf{PP} \sqcup \mathsf{PPI}$. See Wessel [10, Section 4.3]. We therefore work exclusively with strong EQ.

Remark 2.4 (Key structural properties)**.** In any $\mathcal{ALCI}_{RCC5}$ interpretation:

- PP is a strict partial order (irreflexive, asymmetric, transitive).
- There is no finite model property: the concept $(\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ is satisfiable only in infinite models.
- There is no tree model property: all models are complete graphs.
- The *universal role* $R_* = \mathsf{DR} \sqcup \mathsf{PO} \sqcup \mathsf{EQ} \sqcup \mathsf{PP} \sqcup \mathsf{PPI}$ is definable, enabling TBox internalization.
- Under strong EQ, a *difference role* $R_D = R_* \setminus \mathsf{EQ}$ is available, enabling limited nominal expressivity.

2.2 The Patchwork Property of RCC5

We recall the fundamental result from qualitative constraint reasoning that is the lynchpin of our decidability proof.

Definition 2.5 (Atomic constraint network)**.** An *atomic RCC5 constraint network* N over a set of variables V assigns to each ordered pair (v_i, v_j) with $v_i \neq v_j$ a single base relation $R_{ij} \in N_R \setminus \{\mathsf{EQ}\}$, and assigns EQ to each (v_i, v_i), respecting converse ($R_{ji} = \mathsf{inv}(R_{ij})$).

Definition 2.6 (Path consistency)**.** An atomic network N is *path-consistent* if for every triple of variables (v_i, v_j, v_k):

$$R_{ik} \in \mathsf{comp}(R_{ij}, R_{jk}).$$

Definition 2.7 (Consistency)**.** An atomic network N is *consistent* (or *realizable*) if there exists an $\mathcal{ALCI}_{RCC5}$ interpretation $\mathcal{I}$ and elements $d_v \in \Delta^{\mathcal{I}}$ for each $v \in V$ such that $(d_{v_i}, d_{v_j}) \in R_{ij}^{\mathcal{I}}$ for all i, j.

Theorem 2.8 (Renz & Nebel [8]; Renz [7])**.** *For RCC5 (and for RCC8), an atomic constraint network is consistent if and only if it is path-consistent.*

Corollary 2.9 (Patchwork property)**.** *Let N_1 and N_2 be consistent atomic RCC5 networks that agree on their common variables. Then $N_1 \cup N_2$ is consistent.*

Proof. Path consistency is a local (triple-wise) condition. Every triple in $N_1 \cup N_2$ either lies entirely in N_1, entirely in N_2, or involves variables from both—in the last case, the shared variables carry the same labels in both networks, and path consistency of the triple follows from the individual path consistencies and the agreement on shared variables. By Theorem 2.8, $N_1 \cup N_2$ is consistent. □

The patchwork property is the engine of our decidability proof: it ensures that *local (triple-wise) consistency implies global consistency* for atomic RCC5 networks.

3 Monotonicity Along PP-Chains

Before developing the main proof techniques, we establish a structural property of $\mathcal{ALCI}_{RCC5}$ models that constrains the behavior of infinite PP-chains and explains why $\mathcal{ALCI}_{RCC5}$ is "tamer" than one might expect.

Lemma 3.1 (Upward monotonicity)**.** *Let $\mathcal{I}$ be an $\mathcal{ALCI}_{RCC5}$ interpretation containing a PP-chain $a_0, a_1, a_2, \ldots$ (i.e., $(a_i, a_{i+1}) \in \mathsf{PP}^{\mathcal{I}}$ for all i). For any $x \in \Delta^{\mathcal{I}}$ and any $i \geq 0$, the relation $R(x, a_{i+1})$ is constrained by $R(x, a_i)$ as follows:*

$R(x, a_i)$	*Possible values of* $R(x, a_{i+1})$
PP	PP
PO	PO, PP
DR	DR, PO, PP
PPI	PO, EQ, PP, PPI

Proof. Direct from the composition table: $R(x, a_{i+1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PP})$. Reading column $\mathsf{PP}(a, b)$ in Table 1 for each row gives the claimed sets. □

Corollary 3.2. *Going upward along a PP-chain, the relation of an external node progresses monotonically through at most four phases. The "strength" ordering* DR < PO < PP *is respected (with* PPI *on a separate track that merges into* PO/PP*). Reversals* PP → PO*,* PO → DR*, or* PP → DR *are impossible.*

Lemma 3.3 (Downward persistence)**.** *Going* downward *along a PP-chain (i.e., from a_i to a_{i-1} via* PPI*), the relation $R(x, a_{i-1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PPI})$ satisfies:*

$R(x, a_i)$	*Possible values of* $R(x, a_{i-1})$
DR	DR
PPI	PPI
PO	PO, PPI
PP	PO, EQ, PP, PPI

In particular, DR *and* PPI *are downward closed.*

Proof. By $R(x, a_{i-1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PPI})$, reading column $\mathsf{PPI}(a, b)$ from Table 1. □

Remark 3.4. The monotonicity lemmas show that an infinite PP-chain can only "absorb" external nodes in a structured way: once a node becomes DR or PPI relative to a chain element, it stays that way for all elements below; once it becomes PP, it stays PP for all elements above. This regularity is what allows a finite quasimodel to faithfully represent the infinite model structure.

4 The Quasimodel Method

Throughout this section, fix a concept C_0 in negation normal form (NNF).

4.1 Fischer–Ladner Closure

Definition 4.1 (Closure)**.** The *closure* $\mathsf{cl}(C_0)$ is the smallest set containing $\mathsf{sub}(C_0)$ (all subconcepts of C_0) and, for every concept name $A \in N_C$ occurring in C_0, the concept $\neg A$. We identify $\neg\neg D$ with D. Note $|\mathsf{cl}(C_0)| \leq 2|C_0|$.

4.2 Types

Definition 4.2 (Type)**.** A *type* (for C_0) is a subset $\tau \subseteq \mathsf{cl}(C_0)$ satisfying:
(T1) If $D_1 \sqcap D_2 \in \tau$, then $D_1 \in \tau$ and $D_2 \in \tau$.
(T2) If $D_1 \sqcup D_2 \in \tau$, then $D_1 \in \tau$ or $D_2 \in \tau$.
(T3) For every concept name A: not both $A \in \tau$ and $\neg A \in \tau$.
We write $\mathsf{Tp}(C_0)$ for the set of all types. Clearly $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|} \leq 2^{2|C_0|}$.

Given an interpretation $\mathcal{I}$ and $x \in \Delta^{\mathcal{I}}$, the type of x is $\tau^{\mathcal{I}}(x) = \{D \in \mathsf{cl}(C_0) \mid x \in D^{\mathcal{I}}\}$.

4.3 Pair-Types (R-Compatibility)

Definition 4.3 (R-compatibility)**.** Let $\tau_1, \tau_2 \in \mathsf{Tp}(C_0)$ and $R \in N_R \setminus \{\mathsf{EQ}\}$. The triple (τ_1, R, τ_2) is *R-compatible* if:
(P1) For all $\forall R.D \in \mathsf{cl}(C_0)$: if $\forall R.D \in \tau_1$ then $D \in \tau_2$.
(P2) For all $\forall \mathsf{inv}(R).D \in \mathsf{cl}(C_0)$: if $\forall \mathsf{inv}(R).D \in \tau_2$ then $D \in \tau_1$.
(P3) For all $\forall S.D \in \mathsf{cl}(C_0)$ with $S \neq R$ and $S \neq \mathsf{inv}(R)$: no new constraint is imposed by S on this pair. (The pair is related by R, not S.)
We define:

$$\mathsf{Pair}(C_0) = \{(\tau_1, R, \tau_2) \mid \tau_1, \tau_2 \in \mathsf{Tp}(C_0),\ R \in N_R \setminus \{\mathsf{EQ}\},\ (\tau_1, R, \tau_2) \text{ is } R\text{-compatible}\}.$$

Note that since we work with strong EQ semantics, the EQ relation holds only reflexively, and we handle it separately: $(\tau, \mathsf{EQ}, \tau)$ is always implicitly present for any type τ.

Remark 4.4. R-compatibility captures the *local* constraints imposed by universal restrictions. If $(\tau_1, R, \tau_2) \in \mathsf{Pair}(C_0)$, then two domain elements of types τ_1 and τ_2 could consistently be R-related—as far as universal restrictions are concerned.

4.4 Triple Consistency

Definition 4.5 (Consistent triple)**.** A *triple* $(\tau_1, R_{12}, \tau_2, R_{13}, \tau_3, R_{23})$ is *consistent* if:
(Tr1) Each pair is compatible: (τ_1, R_{12}, τ_2), (τ_1, R_{13}, τ_3), (τ_2, R_{23}, τ_3) are all in $\mathsf{Pair}(C_0)$.
(Tr2) The composition table is respected:

$$\begin{aligned} &R_{13} \in \mathsf{comp}(R_{12}, R_{23}), \\ &R_{23} \in \mathsf{comp}(\mathsf{inv}(R_{12}), R_{13}), \\ &R_{12} \in \mathsf{comp}(R_{13}, \mathsf{inv}(R_{23})). \end{aligned}$$

We write $\mathsf{Trip}(C_0)$ for the set of all consistent triples.

4.5 Quasimodel

Definition 4.6 (Quasimodel)**.** A *quasimodel* for C_0 is a triple $\mathcal{Q} = (T, P, \tau_0)$ where:

- $T \subseteq \mathsf{Tp}(C_0)$ is a non-empty set of types,
- $P \subseteq \mathsf{Pair}(C_0)$ is a set of pair-types restricted to T,
- $\tau_0 \in T$ is a distinguished type with $C_0 \in \tau_0$,

satisfying the following conditions:

(Q1) Existential demand satisfaction. For every $\tau \in T$ and every $\exists R.D \in \tau$, there exists a *witness type* $\tau' \in T$ with $D \in \tau'$ and $(\tau, R, \tau') \in P$, such that:

(Q1.a) *Pairwise embeddability with existing types:* For every $\tau'' \in T$ and every $R'' \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau, R'', \tau'') \in P$, there exists $S \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau'', S, \tau') \in P$ and $(\tau, R'', \tau'', R, \tau', S) \in \mathsf{Trip}(C_0)$.

(Q1.b) *Full triple coherence:* For every $\tau_a, \tau_b \in T$ and every R_{ab} with $(\tau_a, R_{ab}, \tau_b) \in P$, there exist $S_a, S_b \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau_a, S_a, \tau') \in P$ and $(\tau_b, S_b, \tau') \in P$ such that $(\tau_a, R_{ab}, \tau_b, S_a, \tau', S_b) \in \mathsf{Trip}(C_0)$.

(Q2) Completeness. For every $\tau_1, \tau_2 \in T$, there exists $R \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau_1, R, \tau_2) \in P$.

(Q3) PP-transitivity. If $(\tau_1, \mathsf{PP}, \tau_2) \in P$ and $(\tau_2, \mathsf{PP}, \tau_3) \in P$, then $(\tau_1, \mathsf{PP}, \tau_3) \in P$, and for all $\forall \mathsf{PP}.D \in \mathsf{cl}(C_0)$: if $\forall \mathsf{PP}.D \in \tau_1$ then $D \in \tau_3$. Symmetrically for PPI.

Remark 4.7. Condition (Q1) is the heart of the quasimodel. It ensures that when we add a new element (to witness an existential demand), we can assign it a role to *every* existing element in a way that is simultaneously consistent. Conditions (Q1.a) and (Q1.b) correspond to different perspectives on this requirement. Condition (Q2) ensures the one-cluster property. Condition (Q3) handles the transitivity of PP (and PPI), which propagates universal restrictions along PP-chains.

5 Main Theorem: Quasimodel Characterization

Theorem 5.1. *A concept C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$ if and only if there exists a quasimodel for C_0.*

The proof occupies the next two subsections.

5.1 Soundness: Model ⇒ Quasimodel

Proof of soundness. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be an $\mathcal{ALCI}_{RCC5}$ interpretation with $x_0 \in C_0^{\mathcal{I}}$. Define:

$$
\begin{aligned}
T &= \{\tau^{\mathcal{I}}(x) \mid x \in \Delta^{\mathcal{I}}\}, \\
P &= \{(\tau^{\mathcal{I}}(x), R, \tau^{\mathcal{I}}(y)) \mid (x, y) \in R^{\mathcal{I}},\ R \in N_R \setminus \{\mathsf{EQ}\}\}, \\
\tau_0 &= \tau^{\mathcal{I}}(x_0).
\end{aligned}
$$

We verify each quasimodel condition:

(Q1) Let $\tau = \tau^{\mathcal{I}}(x)$ and $\exists R.D \in \tau$. Then there exists y with $(x, y) \in R^{\mathcal{I}}$ and $y \in D^{\mathcal{I}}$. Set $\tau' = \tau^{\mathcal{I}}(y)$.

For (Q1.a): given any $\tau'' = \tau^{\mathcal{I}}(z)$ with $(\tau, R'', \tau'') \in P$, witnessed by some z' with $\tau^{\mathcal{I}}(z') = \tau''$ and $(x, z') \in (R'')^{\mathcal{I}}$, the relation S between z' and y in $\mathcal{I}$ satisfies all required conditions by the model properties.

For (Q1.b): given any $\tau_a = \tau^{\mathcal{I}}(a)$, $\tau_b = \tau^{\mathcal{I}}(b)$ with $(\tau_a, R_{ab}, \tau_b) \in P$, witnessed by some a', b' with the appropriate types and relations, the relations of a' and b' to y provide the required S_a, S_b.

(Q2) By the one-cluster property, every pair of elements has a role, so every pair of types has a role in P.

(Q3) Transitivity of $\mathsf{PP}^{\mathcal{I}}$ ensures that if $(\tau_1, \mathsf{PP}, \tau_2) \in P$ and $(\tau_2, \mathsf{PP}, \tau_3) \in P$, then there exist elements x_1, x_2, x_3 such that $(x_1, x_2) \in \mathsf{PP}^{\mathcal{I}}$, $(x_2, x_3) \in \mathsf{PP}^{\mathcal{I}}$, and hence $(x_1, x_3) \in \mathsf{PP}^{\mathcal{I}}$, giving $(\tau_1, \mathsf{PP}, \tau_3) \in P$. The propagation of universal restrictions follows from the model satisfying $\forall \mathsf{PP}.D$. □

5.2 Completeness: Quasimodel ⇒ Model

This is the non-trivial direction.

Proof of completeness. Let $\mathcal{Q} = (T, P, \tau_0)$ be a quasimodel for C_0. We construct a model $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ via a *Henkin-style step-by-step construction.*

Stage 0. Create a single element e_1 with type $\tau(e_1) := \tau_0$. Set $\Delta_0 = \{e_1\}$.

Stage $n+1$. Let $\Delta_n = \{e_1, \ldots, e_m\}$ with types $\tau(e_i)$ and roles $R(e_i, e_j) \in N_R$ assigned for all $i \neq j$ (with $R(e_i, e_i) = \mathsf{EQ}$). Assume inductively:

(i) $(\tau(e_i), R(e_i, e_j), \tau(e_j)) \in P$ for all $i \neq j$.

(ii) Every triple (e_i, e_j, e_k) satisfies the composition table.

Enumerate all unsatisfied demands: pairs $(e_k, \exists R.D)$ where $\exists R.D \in \tau(e_k)$ but no element in Δ_n serves as a witness. Consider the next unsatisfied demand $(e_k, \exists R.D)$.

By condition (Q1) of the quasimodel, there exists a witness type $\tau' \in T$ with $D \in \tau'$ and $(\tau(e_k), R, \tau') \in P$. Create a new element e_{m+1} with $\tau(e_{m+1}) := \tau'$. Set $R(e_k, e_{m+1}) := R$ and $R(e_{m+1}, e_k) := \mathsf{inv}(R)$.

The key step is assigning a role $R(e_i, e_{m+1})$ for every existing element $e_i \in \Delta_n$ with $i \neq k$.

Claim 5.2. *There exist roles $S_1, \ldots, S_m \in N_R \setminus \{\mathsf{EQ}\}$ (with $S_k = R$) such that:*

(a) $(\tau(e_i), S_i, \tau') \in P$ for all i.

(b) For all $i \neq j$: the triple (e_i, e_j, e_{m+1}) with roles $R(e_i, e_j)$, S_i, S_j satisfies $S_i \in \mathsf{comp}(R(e_i, e_j), S_j)$ and is in $\mathsf{Trip}(C_0)$.

Proof of Claim 5.2. Consider the problem of choosing roles $S_1, \ldots, S_m$ as a constraint satisfaction problem. Define an atomic RCC5 constraint network N on the variables $\{e_1, \ldots, e_m, e_{m+1}\}$:

- For $i, j \leq m$: $N(e_i, e_j) = R(e_i, e_j)$ (the existing, consistent network).
- For $i \leq m$: $N(e_i, e_{m+1}) = S_i$ (to be determined), with $S_k = R$ fixed.
- The domain of S_i is $\{S \in N_R \setminus \{\mathsf{EQ}\} \mid (\tau(e_i), S, \tau') \in P\}$.

The constraints on the S_i are: for all $i \neq j$ with $i, j \leq m$,

$$S_i \in \mathsf{comp}(R(e_i, e_j), S_j) \quad \text{and} \quad S_j \in \mathsf{comp}(R(e_j, e_i), S_i).$$

Additionally, the triple $(\tau(e_i), R(e_i, e_j), \tau(e_j), S_i, \tau', S_j)$ must be in $\mathsf{Trip}(C_0)$.

Now, condition (Q1.b) of the quasimodel guarantees that for every *pair* (e_i, e_j), there *exist* values S_i, S_j satisfying these constraints. In other words, every pairwise sub-network $\{e_i, e_j, e_{m+1}\}$ is path-consistent.

We need these pairwise-consistent choices to extend to a *globally* consistent assignment. This is precisely the content of the **patchwork property** (Corollary 2.9).

More explicitly: the constraint network N restricted to any triple $\{e_i, e_j, e_{m+1}\}$ is consistent (by the pairwise argument above), so the full network N is path-consistent. By Theorem 2.8, a path-consistent atomic RCC5 network is globally consistent. Therefore, a simultaneous assignment $S_1, \ldots, S_m$ satisfying all constraints exists. □

Set $R(e_i, e_{m+1}) := S_i$ and $R(e_{m+1}, e_i) := \mathsf{inv}(S_i)$ for all i. This extends the interpretation to $\Delta_{n+1} = \Delta_n \cup \{e_{m+1}\}$ while maintaining the inductive invariants (i) and (ii).

The limit model. Using a fair enumeration of demands (e.g., dovetailing), define the limit interpretation:

$$\mathcal{I} = \bigcup_{n \geq 0} (\Delta_n, \cdot_n^{\mathcal{I}}).$$

This is a valid $\mathcal{ALCI}_{RCC5}$ interpretation:

- *One cluster:* every pair of elements created in the construction has a role assigned.
- *JEPD:* each pair has exactly one base relation (by construction).
- *Composition:* every triple satisfies the composition table (maintained at each stage by the claim).
- *Converse:* maintained by setting $R(e_{m+1}, e_i) = \mathsf{inv}(S_i)$.
- *Strong EQ:* $R(e_i, e_i) = \mathsf{EQ}$ for all i, and no distinct pair has EQ.
- *Concept truth:* for each type τ, every element e with $\tau(e) = \tau$ satisfies exactly the concept descriptions in τ. Since $C_0 \in \tau_0 = \tau(e_1)$, we have $e_1 \in C_0^{\mathcal{I}}$.

More precisely, concept truth is established by induction on concept structure:

- Concept names and negations: by the type conditions (T1)–(T3).
- Conjunctions, disjunctions: by (T1), (T2).
- Existential restrictions $\exists R.D$: every demand $\exists R.D \in \tau(e)$ is eventually satisfied by the Henkin construction (fair enumeration).
- Universal restrictions $\forall R.D$: if $\forall R.D \in \tau(e)$ and $(e, e') \in R^{\mathcal{I}}$, then $D \in \tau(e')$ by the R-compatibility condition (P1) in $\mathsf{Pair}(C_0)$.

Therefore $\mathcal{I} \models C_0$, and C_0 is satisfiable. □

6 Decidability and Complexity via Type Elimination

Theorem 6.1. *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable in* ExpTime.

Proof. By Theorem 5.1, it suffices to decide quasimodel existence. We describe a type-elimination algorithm.

1. **Initialization.** Compute $\mathsf{Tp}(C_0)$, $\mathsf{Pair}(C_0)$, and $\mathsf{Trip}(C_0)$. These sets have sizes bounded by $2^{O(|C_0|)}$, $2^{O(|C_0|)}$, and $2^{O(|C_0|)}$ respectively.
2. **PP-transitivity closure.** Close $\mathsf{Pair}(C_0)$ under condition (Q3): iteratively add $(\tau_1, \mathsf{PP}, \tau_3)$ whenever $(\tau_1, \mathsf{PP}, \tau_2)$ and $(\tau_2, \mathsf{PP}, \tau_3)$ are present and the universal restriction propagation is satisfied. Remove triples that violate (Q3).
3. **Type elimination.** Iteratively remove from T (initialized to $\mathsf{Tp}(C_0)$) any type τ that fails one of:
   - *Existential demand:* there exists $\exists R.D \in \tau$ for which no witness τ' satisfying (Q1) can be found.
   - *Completeness:* there exists $\tau' \in T$ such that no role R yields $(\tau, R, \tau') \in P$.

   When a type is removed, also remove all pair-types and triple-types involving it, then repeat.
4. **Termination.** Each iteration removes at least one type or pair-type. Since there are at most $2^{O(|C_0|)}$ types and $2^{O(|C_0|)}$ pair-types, termination occurs after at most $2^{O(|C_0|)}$ iterations.
5. **Acceptance.** Accept if and only if there remains a type τ_0 with $C_0 \in \tau_0$ such that τ_0 and the surviving types and pair-types form a quasimodel.

Each iteration inspects at most $2^{O(|C_0|)}$ types, pairs, and triples. Total time: $2^{O(|C_0|)} \cdot 2^{O(|C_0|)} = 2^{O(|C_0|)}$, which is in ExpTime. □

Theorem 6.2 (Lower bound)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is PSpace-hard.*

Proof. Wessel [10, Section 6.3] establishes this by a reduction from QBF validity, using the PO-free fragment to enforce tree-shaped models. □

Corollary 6.3. *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is between PSpace and ExpTime.*

Remark 6.4 (Closing the gap)**.** We conjecture that $\mathcal{ALCI}_{RCC5}$ is ExpTime-complete. The interaction between PP-transitivity and the universal role (which enables TBox internalization) plausibly supports a reduction from the ExpTime-hard *two-player corridor tiling problem*, but we have not established this.

7 A Tableau Decision Procedure

We now present an alternative decision procedure in the form of a *tableau calculus with blocking*. Unlike the type-elimination algorithm, which works globally on the space of all types and pair-types, the tableau constructs a model-like structure incrementally, expanding individual nodes. The key challenge is that $\mathcal{ALCI}_{RCC5}$ models are complete graphs, not trees: creating a new node requires assigning it a base relation to *every* existing node. The patchwork property (Theorem 2.8) is what makes this tractable.

7.1 Completion Graphs

Definition 7.1 (Completion graph)**.** A *completion graph* for C_0 is a tuple $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ where:

- V is a finite, non-empty set of *nodes*.
- $\mathcal{L}\colon V \to 2^{\mathsf{cl}(C_0)}$ is a *labelling function* assigning a set of concepts to each node.
- $\mathcal{E}\colon \{(x, y) \in V \times V \mid x \neq y\} \to N_R \setminus \{\mathsf{EQ}\}$ is an *edge labelling function* assigning a base relation to every ordered pair of distinct nodes, respecting converse: $\mathcal{E}(y, x) = \mathsf{inv}(\mathcal{E}(x, y))$.
- $\prec$ is a strict total order on V (the *creation order*).

The *initial completion graph* for C_0 consists of a single node x_0 with $\mathcal{L}(x_0) = \{C_0\}$, $V = \{x_0\}$, and $\mathcal{E}$ vacuously defined (no pairs of distinct nodes yet).

Remark 7.2. The edge labelling $\mathcal{E}$ is a *total* function on distinct pairs, reflecting the one-cluster and JEPD requirements. Whenever a new node y is added to V, the function $\mathcal{E}$ must be extended to assign a base relation $\mathcal{E}(z, y)$ for every existing node z. This is the principal difference from tree-based tableau calculi, where a new node has a single edge to its parent.

Definition 7.3 (Blocking)**.** Let $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ be a completion graph. A node $x \in V$ is *blocked* if there exists $y \prec x$ such that y is not blocked and $\mathcal{L}(x) = \mathcal{L}(y)$. We call y a *blocker* for x. A node is *active* if it is not blocked.

Remark 7.4. This is *equality anywhere-blocking*: the blocker y need not be a predecessor or neighbor in any structural sense—it need only precede x in the creation order and carry the same label. The simplicity of this blocking condition (compared to, e.g., the pairwise blocking of $\mathcal{ALCI}$ tree tableaux) is made possible by the patchwork property, which allows us to decouple the type-level structure from the specific relational neighborhood (see the soundness proof, Section 7.6).

7.2 Expansion Rules

The tableau calculus consists of four deterministic rules and two nondeterministic rules, applied to a completion graph $\mathcal{G}$. We write $\mathcal{L}(x)$ for the label and $\mathcal{E}(x, y)$ for the edge label.

Propositional rules (deterministic):

($\sqcap$) If $D_1 \sqcap D_2 \in \mathcal{L}(x)$ and $\{D_1, D_2\} \not\subseteq \mathcal{L}(x)$:
add D_1 and D_2 to $\mathcal{L}(x)$.

($\sqcup$) If $D_1 \sqcup D_2 \in \mathcal{L}(x)$ and $D_1 \notin \mathcal{L}(x)$ and $D_2 \notin \mathcal{L}(x)$:
nondeterministically add D_1 or D_2 to $\mathcal{L}(x)$.

Propagation rule (deterministic):

($\forall$) If $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = R$ and $D \notin \mathcal{L}(y)$:
add D to $\mathcal{L}(y)$.

This rule also handles inverses: since $\forall R^{-}.D$ is equivalent to $\forall \mathsf{inv}(R).D$ in NNF, and $\mathcal{E}$ stores base relations, the rule fires whenever $\forall S.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = S$.

Generating rule (nondeterministic):

($\exists$) If $\exists R.D \in \mathcal{L}(x)$, x is active, and there is no $y \in V$ with $\mathcal{E}(x, y) = R$ and $D \in \mathcal{L}(y)$:

1. Create a fresh node y and add it to V as the $\prec$-greatest element.
2. Set $\mathcal{L}(y) := \{D\}$.
3. Set $\mathcal{E}(x, y) := R$ and $\mathcal{E}(y, x) := \mathsf{inv}(R)$.
4. For each $z \in V \setminus \{x, y\}$, *nondeterministically* choose $S_z \in N_R \setminus \{\mathsf{EQ}\}$ and set $\mathcal{E}(z, y) := S_z$, $\mathcal{E}(y, z) := \mathsf{inv}(S_z)$.

Remark 7.5. Step 4 is the hallmark of the complete-graph tableau: the new node y receives an edge to *every* existing node. The choice of S_z is nondeterministic, subject to the constraint-filtering condition below. The patchwork property guarantees that a valid choice exists whenever the concept is satisfiable (see Section 7.7).

7.3 Constraint Filtering

After the ($\exists$)-rule creates a new node y with edges to all existing nodes, we require:

(CF) Composition consistency. For every triple of distinct nodes (u, v, w) that includes y:

$$\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v),\, \mathcal{E}(v, w)).$$

If no assignment of the S_z values in step 4 of ($\exists$) satisfies (CF) for all such triples simultaneously, the current branch *closes* (clash by composition failure).

Remark 7.6. Checking (CF) amounts to solving an atomic RCC5 constraint network: the existing nodes have fixed pairwise relations, and the new node y has the relation to x fixed at R, with the remaining relations S_z as variables. By Theorem 2.8, a simultaneous satisfying assignment exists if and only if every triple involving y is individually satisfiable, i.e., the network is path-consistent. Path consistency can be checked (and enforced) in time $O(|V|^3)$.

7.4 Rule Application Strategy

To ensure that labels stabilize before blocking is evaluated:

1. Apply ($\sqcap$), ($\sqcup$), and ($\forall$) exhaustively to all nodes (in any order, resolving ($\sqcup$) nondeterministically).
2. Re-evaluate blocking: a node x is blocked if $\mathcal{L}(x) = \mathcal{L}(y)$ for some non-blocked $y \prec x$.
3. Apply one instance of ($\exists$) for an active node with an unsatisfied demand.
4. After adding the new node and its edges, apply ($\forall$) exhaustively (the new edges may trigger propagations both to and from the new node; these may cascade to other nodes).
5. Re-evaluate blocking (label changes from ($\forall$) may affect blocking status).
6. Repeat from step 1 until no rule applies or a clash is detected.

Definition 7.7 (Saturated completion graph)**.** A completion graph $\mathcal{G}$ is *saturated* if no expansion rule is applicable (i.e., all propositional rules have been applied, all ($\forall$)-propagations have been performed, and every active node's existential demands are witnessed by an appropriate neighbor).

Definition 7.8 (Clash-free completion graph)**.** A completion graph is *clash-free* if:
(C1) No node has $\{A, \neg A\} \subseteq \mathcal{L}(x)$ for any concept name A.
(C2) Composition consistency (CF) holds for all triples.
A completion graph that is both saturated and clash-free is called *open*.

The tableau *decides* satisfiability: C_0 is satisfiable iff there exists a sequence of nondeterministic choices producing an open completion graph.

7.5 Termination

Theorem 7.9 (Termination)**.** *Every sequence of rule applications terminates. The final completion graph has at most* $2^{O(|C_0|)}$ *nodes.*

Proof. Let $n = |\mathsf{cl}(C_0)| \leq 2|C_0|$ and $N = 2^n = |\mathsf{Tp}(C_0)|$.

Bound on active nodes. At any point, the active (non-blocked) nodes have pairwise distinct labels (by the definition of blocking). Each label is a subset of $\mathsf{cl}(C_0)$, so there are at most N active nodes.

Labels grow monotonically. The rules ($\sqcap$), ($\sqcup$), and ($\forall$) only *add* concepts to labels, never remove them. Therefore, each node's label changes at most n times (once per concept in $\mathsf{cl}(C_0)$).

Bound on node creation. The ($\exists$) rule fires only for active nodes with unsatisfied demands. Each active node x has at most n existential demands (concepts of the form $\exists R.D$ in $\mathsf{cl}(C_0)$). Each demand generates at most one new node (once a witness exists, the demand is satisfied and the rule no longer fires for that demand). With at most N active nodes, at most $N \cdot n$ nodes are created by the ($\exists$) rule. Each newly created node is either active (contributing to the at-most-N budget) or immediately blocked.

Bound on re-evaluations. When a label changes (via ($\forall$) propagation), blocking status may change, and a previously blocked node may become active, requiring expansion. However, each label change is permanent (labels only grow), and the total number of label changes across all nodes is at most $|V| \cdot n$. Each unblocking event is caused by a label change, so the total number of unblocking events is bounded. Each unblocking event may generate at most n new witnesses, but the total number of witnesses is bounded by the total number of active-node $\times$ demand pairs encountered during the process, which is at most $N \cdot n$ per "generation" of the label.

Overall bound. The total number of nodes is bounded by $N \cdot n \cdot n = 2^n \cdot n^2 = 2^{O(|C_0|)}$, and the total number of rule applications (label additions and node creations) is bounded by $|V| \cdot n = 2^{O(|C_0|)}$. Hence the procedure terminates in at most $2^{O(|C_0|)}$ steps. □

7.6 Soundness: Open Branch ⇒ Satisfiable

Theorem 7.10 (Soundness of the tableau)**.** *If the tableau produces an open* (*saturated, clash-free*) *completion graph* $\mathcal{G}$ *for* C_0*, then* C_0 *is satisfiable in* $\mathcal{ALCI}_{RCC5}$*.*

Proof. We show that an open completion graph $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ encodes a valid quasimodel. The result then follows from Theorem 5.1.

Step 1: Quasimodel extraction. Define:

$$
\begin{aligned}
T &= \{\mathcal{L}(x) \mid x \in V,\ x \text{ is active}\},\\
P &= \{(\mathcal{L}(x), \mathcal{E}(x,y), \mathcal{L}(y)) \mid x, y \in V,\ x \neq y\},\\
\tau_0 &= \mathcal{L}(x_0).
\end{aligned}
$$

Step 2: Verification of (Q1). Let $\tau \in T$ and $\exists R.D \in \tau$. Then $\tau = \mathcal{L}(x)$ for some active node x. Since $\mathcal{G}$ is saturated, there exists $y \in V$ with $\mathcal{E}(x,y) = R$ and $D \in \mathcal{L}(y)$. Set $\tau' = \mathcal{L}(y)$.

For **(Q1.a)**: Let $(\tau, R'', \tau'') \in P$. Then there exist nodes x', z' with $\mathcal{L}(x') = \tau$, $\mathcal{L}(z') = \tau''$, and $\mathcal{E}(x', z') = R''$. Since $\mathcal{G}$ is a complete graph on V, the node z' has an edge to y: set $S = \mathcal{E}(z', y)$. Then $(\tau'', S, \tau') \in P$ by definition. Composition consistency of the triple (x, z', y) with roles (R'', R, S) follows from clash-freeness (C2).

But we need the triple with respect to the node x' (which satisfies $\mathcal{L}(x') = \tau$), not x. If $x' = x$, we are done. If $x' \neq x$, we use the triple (x', z', y): $\mathcal{E}(x', z') = R''$, $\mathcal{E}(x', y) =$ some role R'_0, and $\mathcal{E}(z', y) = S$. Composition consistency holds by (C2).

However, R'_0 might not equal R. The condition (Q1.a) requires a witness relative to the specific role R from τ to τ'. To handle this: note that $\mathcal{L}(x) = \tau$ and $\mathcal{E}(x, y) = R$. For any z with $\mathcal{L}(z) = \tau''$ and $\mathcal{E}(x, z) = R''$, the triple (x, z, y) has roles $(R'', R, \mathcal{E}(z, y))$, and composition holds by (C2). Setting $S = \mathcal{E}(z, y)$ gives $(\tau'', S, \tau') \in P$ and the triple is consistent. This verifies (Q1.a) using the specific node x.

For **(Q1.b)**: Let $(\tau_a, R_{ab}, \tau_b) \in P$ with witnesses $a, b \in V$ such that $\mathcal{L}(a) = \tau_a$, $\mathcal{L}(b) = \tau_b$, $\mathcal{E}(a, b) = R_{ab}$. Set $S_a = \mathcal{E}(a, y)$ and $S_b = \mathcal{E}(b, y)$. Then $(\tau_a, S_a, \tau') \in P$ and $(\tau_b, S_b, \tau') \in P$ by definition. The triple (a, b, y) with roles (R_{ab}, S_a, S_b) is composition-consistent by (C2), so it lies in $\mathsf{Trip}(C_0)$.

Step 3: Verification of (Q2). Since $\mathcal{E}$ is a total function on distinct pairs, for any $\tau_1, \tau_2 \in T$ with active witnesses x_1, x_2, the edge $\mathcal{E}(x_1, x_2)$ provides the required role.

Step 4: Verification of (Q3). Suppose $(\tau_1, \mathsf{PP}, \tau_2) \in P$ and $(\tau_2, \mathsf{PP}, \tau_3) \in P$. Then there exist nodes a, b, c, d with $\mathcal{L}(a) = \tau_1$, $\mathcal{L}(b) = \tau_2 = \mathcal{L}(c)$, $\mathcal{L}(d) = \tau_3$, $\mathcal{E}(a, b) = \mathsf{PP}$, $\mathcal{E}(c, d) = \mathsf{PP}$.

If $b = c$, then the triple (a, b, d) gives $\mathcal{E}(a, d) \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $(\tau_1, \mathsf{PP}, \tau_3) \in P$ and the $(\forall)$-rule ensures universal restriction propagation.

If $b \neq c$, we have $\mathcal{L}(b) = \mathcal{L}(c) = \tau_2$. Consider the edge $\mathcal{E}(a, c)$. Since $\mathcal{L}(a) = \tau_1$ and $\mathcal{L}(c) = \tau_2$, the pair (a, c) contributes $(\tau_1, \mathcal{E}(a, c), \tau_2)$ to P. Now consider the triple (a, c, d): $\mathcal{E}(c, d) = \mathsf{PP}$, so $\mathcal{E}(a, d) \in \mathsf{comp}(\mathcal{E}(a, c), \mathsf{PP})$. We need $(\tau_1, \mathsf{PP}, \tau_3) \in P$. This holds if there exist nodes a', d' with types τ_1, τ_3 and $\mathcal{E}(a', d') = \mathsf{PP}$. We can take $a' = a$ and $d' = d$ provided $\mathcal{E}(a, d) = \mathsf{PP}$. But $\mathcal{E}(a, d)$ might not be PP.

To handle this: we observe that $\mathcal{E}(a, b) = \mathsf{PP}$ and $\mathcal{L}(b) = \mathcal{L}(c)$. We use the saturation of the graph more carefully: if $\mathcal{E}(a, c) = \mathsf{PP}$ (possible since b and c have the same type), then the triple (a, c, d) gives $\mathcal{E}(a, d) \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, and we are done.

In general, P is defined as the set of *all* pair-types realized in $\mathcal{G}$, so it may contain $(\tau_1, \mathsf{PP}, \tau_2)$ from the pair (a, b) and $(\tau_2, \mathsf{PP}, \tau_3)$ from (c, d) without the transitive pair $(\tau_1, \mathsf{PP}, \tau_3)$ being directly realized. However, the $(\forall)$-rule ensures that if $\forall \mathsf{PP}.D \in \mathcal{L}(a)$ and $\mathcal{E}(a, b) = \mathsf{PP}$, then $D \in \mathcal{L}(b) = \tau_2$; and if D is $\forall \mathsf{PP}.D'$, then $D' \in \mathcal{L}(d)$. The universal restriction propagation required by (Q3) follows from the saturation of $(\forall)$. For the pair-type itself, we note that $(\tau_1, \mathsf{PP}, \tau_3)$ will be present in P via the pair (a, d) with $\mathcal{E}(a, d) = \mathsf{PP}$ if a is PP-related to d. If $\mathcal{E}(a, d) \neq \mathsf{PP}$, we can still recover: the quasimodel condition (Q3) only requires $(\tau_1, \mathsf{PP}, \tau_3) \in P$ when both $(\tau_1, \mathsf{PP}, \tau_2)$ and $(\tau_2, \mathsf{PP}, \tau_3)$ are in P. We add the following *closure rule*:

(PP-close) If $\mathcal{E}(x, y) = \mathsf{PP}$ and $\mathcal{E}(y, z) = \mathsf{PP}$ for nodes $x, y, z \in V$, then $\mathcal{E}(x, z)$ must equal PP (otherwise: clash, since $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$).

This is already subsumed by the composition consistency condition (CF). Therefore, in any clash-free completion graph, PP-transitivity holds at the node level, which directly ensures (Q3) at the type level.

Step 5: Conclusion. The extracted triple (T, P, τ_0) is a quasimodel for C_0. By Theorem 5.1, C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$. □

7.7 Completeness: Satisfiable ⇒ Open Branch

Theorem 7.11 (Completeness of the tableau)**.** *If C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$, then there exists a sequence of nondeterministic choices yielding an open completion graph for C_0.*

Proof. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be a model with $d_0 \in C_0^{\mathcal{I}}$. We show that nondeterministic choices can be guided by $\mathcal{I}$ so that no clash arises and every demand is satisfied.

Guide function. We maintain a partial function $\pi\colon V \to \Delta^{\mathcal{I}}$ that maps each tableau node to a "corresponding" element in the model, with the invariant:

(I) $\mathcal{L}(x) \subseteq \tau^{\mathcal{I}}(\pi(x))$ for all $x \in V$.

(II) $\mathcal{E}(x, y) = R^{\mathcal{I}}(\pi(x), \pi(y))$ for all distinct $x, y \in V$ (where $R^{\mathcal{I}}(a, b)$ denotes the unique base relation holding between a and b in $\mathcal{I}$).

Initialization. Set $\pi(x_0) = d_0$. Then $\mathcal{L}(x_0) = \{C_0\}$ and $d_0 \in C_0^{\mathcal{I}}$, so invariant (I) holds.

Guiding the ($\sqcup$) rule. When $D_1 \sqcup D_2 \in \mathcal{L}(x)$, invariant (I) gives $D_1 \sqcup D_2 \in \tau^{\mathcal{I}}(\pi(x))$, so $\pi(x) \in (D_1 \sqcup D_2)^{\mathcal{I}}$, meaning $\pi(x) \in D_1^{\mathcal{I}}$ or $\pi(x) \in D_2^{\mathcal{I}}$. Choose the true disjunct. Invariant (I) is maintained.

Guiding the ($\exists$) rule. When $\exists R.D \in \mathcal{L}(x)$ and x is active, invariant (I) gives $\pi(x) \in (\exists R.D)^{\mathcal{I}}$, so there exists $d' \in \Delta^{\mathcal{I}}$ with $(\pi(x), d') \in R^{\mathcal{I}}$ and $d' \in D^{\mathcal{I}}$.

Create node y and set $\pi(y) = d'$. Then:

- $\mathcal{E}(x, y) = R = R^{\mathcal{I}}(\pi(x), \pi(y))$. Invariant (II) holds for the pair (x, y).
- For each existing node z, choose $S_z = R^{\mathcal{I}}(\pi(z), \pi(y))$ (the actual relation in the model). Set $\mathcal{E}(z, y) = S_z$. Invariant (II) holds for (z, y).
- Composition consistency (CF) holds because the model satisfies the composition table.

After ($\forall$)-propagation, every concept added to $\mathcal{L}(y)$ is of the form D' where $\forall S.D' \in \mathcal{L}(z)$ and $\mathcal{E}(z, y) = S$. By invariant (I), $\forall S.D' \in \tau^{\mathcal{I}}(\pi(z))$, so $(\pi(z), \pi(y)) \in S^{\mathcal{I}}$ implies $\pi(y) \in (D')^{\mathcal{I}}$, i.e., $D' \in \tau^{\mathcal{I}}(\pi(y))$. Similarly for propagation from y back to existing nodes. Hence invariant (I) is maintained for all affected labels.

Clash-freeness. Invariant (I) implies $\mathcal{L}(x) \subseteq \tau^{\mathcal{I}}(\pi(x))$, which is a consistent type, so $\{A, \neg A\} \not\subseteq \mathcal{L}(x)$. Invariant (II) implies $\mathcal{E}(x, y) = R^{\mathcal{I}}(\pi(x), \pi(y))$, and the model satisfies composition, so (CF) holds.

Demand satisfaction despite blocking. The guide function shows that all nondeterministic choices can be made consistently. The only remaining concern is that a blocked node x might have an unsatisfied demand.

Let x be blocked by y (so $\mathcal{L}(x) = \mathcal{L}(y)$ and y is active). Suppose $\exists R.D \in \mathcal{L}(x)$. Then $\exists R.D \in \mathcal{L}(y)$, and since y is active and the graph is saturated, there exists a node w with $\mathcal{E}(y, w) = R$ and $D \in \mathcal{L}(w)$.

The demand of x is $\exists R.D$. We need an R-neighbor of x containing D. Node w is an $\mathcal{E}(y, w)$-neighbor of y with role R, but $\mathcal{E}(x, w)$ might be a different role.

This does not cause a problem *for the tableau*: the tableau does not require blocked nodes' demands to be directly satisfied in the completion graph. The completion graph is not itself a model; it is a finite representation from which a model is constructed (via the quasimodel extraction and Henkin construction in Theorem 7.10). The quasimodel extracted in the soundness proof has the type $\mathcal{L}(x) = \mathcal{L}(y) = \tau \in T$, and the demand $\exists R.D$ of τ is witnessed by the type $\mathcal{L}(w) = \tau'$ via the pair-type $(\tau, R, \tau') \in P$. The Henkin construction produces a model satisfying all demands of all types, including those of blocked nodes.

The existence of the guided sequence of choices shows that the nondeterministic tableau has at least one execution producing a saturated, clash-free completion graph. □

7.8 Summary of the Tableau Procedure

Corollary 7.12. *The tableau calculus of Section 7.2, with blocking (Definition 7.3) and constraint filtering (Section 7.3), is a decision procedure for $\mathcal{ALCI}_{RCC5}$ concept satisfiability:*

- *It* ***terminates*** *on every input (Theorem 7.9).*
- *C_0 is satisfiable iff there exists a sequence of choices yielding an open completion graph.*

- ***Soundness**: an open graph implies satisfiability (Theorem 7.10).*
- ***Completeness**: satisfiability implies an open graph exists (Theorem 7.11).*

The worst-case complexity is ExpTime, matching the type-elimination procedure of Section 6.

Remark 7.13 (Comparison with tree-based tableaux)**.** In standard $\mathcal{ALCI}$ tableaux, the completion structure is a forest, new nodes have a single edge to their parent, and blocking uses *pairwise* matching (comparing both a node and its parent with the blocker and its parent) to handle inverse roles. In the $\mathcal{ALCI}_{RCC5}$ tableau:

1. The completion structure is a *complete graph*, not a tree.
2. The ($\exists$)-rule creates edges to *all* existing nodes, not just one.
3. Composition consistency (CF) replaces the tree-structure invariant.
4. Blocking is *simpler*—just type equality, with no need for pairwise matching—because the patchwork property handles the relational context: the quasimodel extraction (Theorem 7.10) shows that type-level information suffices to guarantee model existence, regardless of the specific role assignments between nodes.
5. Soundness uses an indirect argument (quasimodel extraction + Henkin construction) rather than directly reading off the model from the completion graph.

Remark 7.14 (Practical optimizations)**.** An implementation of this tableau can benefit from:

- *Lazy role assignment*: Instead of choosing all $\mathcal{E}(z, y)$ at node creation, maintain a constraint network and only instantiate roles when needed for ($\forall$)-propagation.
- *Path consistency preprocessing*: Before node creation, run arc consistency on the constraint network to prune impossible role values.
- *Dependency-directed backtracking*: When a clash is detected, track which ($\sqcup$) and ($\exists$)-choices contributed, and backtrack to the most recent relevant choice.
- *Caching*: Since blocking is type-based, previously explored subtrees for a given type can be reused.

8 Extension to $\mathcal{ALCI}_{RCC8}$

The RCC8 calculus uses eight base relations: DC, EC, PO, EQ, TPP, TPPI, NTPP, NTPPI. The logic $\mathcal{ALCI}_{RCC8}$ is defined analogously to $\mathcal{ALCI}_{RCC5}$, using the RCC8 composition table.

Theorem 8.1. *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is decidable in ExpTime.*

Proof. Both the type-elimination and tableau procedures extend to $\mathcal{ALCI}_{RCC8}$. The crucial ingredient is that RCC8 also enjoys the patchwork property: path-consistent atomic RCC8 networks are globally consistent (Renz [7]). Therefore:

- Types, pair-types, and triple-types are defined over the RCC8 base relations.
- The quasimodel and tableau conditions use the RCC8 composition table.
- The Henkin construction (Claim 5.2) and the tableau's constraint filtering use the RCC8 patchwork property.
- The type-elimination algorithm runs in $2^{O(|C_0|)}$ time.
- The tableau has at most $2^{O(|C_0|)}$ nodes (7 non-EQ base relations instead of 4, but this is a constant factor).

The soundness and completeness arguments carry over verbatim. □

Theorem 8.2 (Wessel [10])**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is ExpTime-hard.*

Proof. By reduction from the two-player corridor tiling game [10, Section 6.4]. □

Corollary 8.3. *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is ExpTime-complete.*

9 Related Work

Wessel's $\mathcal{ALCRA}_{SG}$ (2000/2001). The $\mathcal{ALCRA}_{SG}$ framework [9] handles role boxes that are *admissible*: deterministic (each composition yields a single relation), functional, complete, and associative. For such role boxes, the logic reduces to $\mathcal{ALC}$ with general TBoxes, yielding ExpTime-completeness. The RCC5 and RCC8 role boxes are non-deterministic (many composition entries yield sets of relations) and hence fall outside $\mathcal{ALCRA}_{SG}$. Our quasimodel and tableau methods handle this non-determinism by working with all possible completions simultaneously, guided by the patchwork property.

Lutz & Wolter (2006): Modal logics of topological relations. Lutz and Wolter [5] study modal logics with RCC8 modalities interpreted over *actual topological spaces*. In their setting, models must correspond to genuine spatial configurations (e.g., in $\mathbb{R}^n$), which imposes additional constraints beyond the composition table. This leads to undecidability in many cases. By contrast, $\mathcal{ALCI}_{RCC}$ uses *abstract* semantics: any interpretation satisfying the composition table constraints is admissible. The abstract semantics are more permissive, and this permissiveness is exactly what makes our decidability proof work.

Lutz & Miličić (2007): ω-admissible concrete domains. Lutz and Miličić [4] prove that $\mathcal{ALC}$ extended with ω-admissible concrete domains is decidable. RCC5 and RCC8 are ω-admissible, so $\mathcal{ALC}$(RCC5) and $\mathcal{ALC}$(RCC8) are decidable. However, the concrete-domain approach is fundamentally different: spatial predicates constrain *concrete features* of domain elements via functional roles, and one cannot quantify *over* spatial relations. In particular, $\forall \mathsf{PP}.C$ and $\exists \mathsf{DR}.D$ are not expressible in $\mathcal{ALC}$(RCC5). The $\mathcal{ALCI}_{RCC}$ logics are strictly more expressive for spatial reasoning because spatial relations serve directly as roles.

Borgwardt, De Bortoli & Koopmann (2024). Borgwardt et al. [2] sharpen the complexity of reasoning in $\mathcal{ALC}$ with ω-admissible concrete domains. Their results apply to the concrete-domain setting and do not directly bear on $\mathcal{ALCI}_{RCC}$, but confirm that the patchwork/amalgamation property of RCC calculi is a powerful tool for decidability in description logics with spatial features.

Constraint satisfaction and qualitative reasoning. The patchwork property (Theorem 2.8) was established by Renz and Nebel [8] and Renz [7] in the context of qualitative spatial reasoning and constraint satisfaction. Our contribution is to bridge the gap between this CSP result and the description logic setting: we show that the patchwork property, when combined with the quasimodel method or a tableau calculus, is precisely the tool needed to decide satisfiability in $\mathcal{ALCI}_{RCC}$ logics.

10 Discussion and Open Questions

1. **Exact complexity of $\mathcal{ALCI}_{RCC5}$.** We have established PSpace $\leq \mathcal{ALCI}_{RCC5} \leq$ ExpTime. The PSpace lower bound [10] uses only the PO-free fragment. We conjecture that the full logic is ExpTime-complete, but closing this gap requires a new lower-bound argument.

2. **General characterization.** Which composition tables (beyond RCC5 and RCC8) yield decidable $\mathcal{ALCI}_{RCC}$-style logics? Our proof shows that the patchwork property is sufficient. It would be valuable to characterize exactly which qualitative calculi have this property and whether it is also necessary for decidability.

3. **Adding number restrictions.** Unqualified number restrictions make $\mathcal{ALCRA}_{SG}$ undecidable [9]. We expect the same for $\mathcal{ALCI}_{RCC5}$: number restrictions would allow enforcing specific branching patterns that, combined with concept distinctions, may enable grid encodings.

4. **Finite model reasoning.** Satisfiability restricted to *finite* models (ruling out infinite PP-chains) may be decidable in a lower complexity class, since the need for infinite witnesses—and hence the full power of the Henkin construction—disappears.

5. **Hybrid extensions.** Wessel [10] considers hybrid logic extensions with nominals and the @-operator. Whether our decidability results extend to these enrichments is an open question; the interaction between nominals and the complete-graph semantics may introduce new complications.

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